

Answer to the question no: 01

A.

Yield: The percentage of good dies from the total number of dies of a wafer.

$$\gamma = \frac{1}{\left(1 + \frac{AD_0}{N}\right)^N}$$

where,

A = die area (cm^2)

D_0 = defects per cm^2 .

N = clustering parameters.

B.

Dies per wafer $\approx \frac{\text{Wafer area}}{\text{Die area}}$ is an approximation because:-

(i) Dies are rectangular and can not tile the circular wafer perfectly.

Partial dies at the wafer edge are discarded

(ii) Spacing between dies and alignment margins that reduce usable area.

So, water area ÷ die area gives the theoretical maximum, the actual number of dies are smaller

C. For water X:

$$\text{diameter} = 19 \text{ cm}$$

$$\therefore \text{radius} = 9.5 \text{ cm}$$

$$\therefore \text{water area} = \pi r^2$$

$$= 283.528 \text{ cm}^2$$

again,

$$\text{Die area } A = \frac{\text{water area}}{\text{dies per water}}$$

$$= \frac{283.528}{69} = 3.1857$$

$$\therefore \text{Yield } Y = \frac{1}{\left(1 + \frac{AD_0}{N}\right)^N}$$

$$= \frac{1}{\left(1 + \frac{3.1857 \times 0.023}{2}\right)^2}$$

$$= 0.9366$$

$$= 93.66\%$$

For water γ :

radius $r = 10 \text{ cm}$

$$\begin{aligned}\therefore \text{area of water} &= \pi r^2 \\ &= \pi \times 10^2 \\ &= 314.15\end{aligned}$$

again,

$$\text{The area } A = \frac{\text{water area}}{\text{dies per water}}$$

$$\begin{aligned}&= \frac{314.15}{100} \\ &= 3.1415\end{aligned}$$

$$\begin{aligned}\therefore \text{Yield} &= \frac{1}{\left(1 + \frac{AD_0}{N}\right)^N} \\ &= \frac{1}{\left(1 + \frac{3.1415 \times 0.031}{2}\right)^2}\end{aligned}$$

$$= 0.9093 = 90.93\%$$

(D) For water x:

$$\text{Water cost} = 20$$

$$\text{Total dies} = 89$$

$$\text{Yield} = 0.9306$$

$$\begin{aligned}\therefore \text{cost per die} &= \frac{\text{water cost}}{\text{dies per water} \times \text{Yield}} \\ &= \frac{20}{89 \times 0.9306} \\ &= 0.24149 \text{ per good die}\end{aligned}$$

For water y:

$$\text{water cost} = 15$$

$$\text{dies per water} = 100$$

$$\text{Yield} = 0.9092$$

$$\begin{aligned}\therefore \text{cost per die} &= \frac{\text{water cost}}{\text{dies per water} \times \text{Yield}} \\ &= \frac{15}{100 \times 0.9092} \\ &= 0.16496 \text{ per good die.}\end{aligned}$$

Answer to the question no: 3

(a) Given that,

P1: clock rate = 3GHz = 3×10^9 Hz, CPI = 1.5

P2: clock rate = 2.5GHz = 2.5×10^9 Hz, CPI = 1

P3: clock rate = 4GHz = 4×10^9 Hz, CPI = 2.2

$\therefore$ Instruction per second = $\frac{\text{Clock rate}}{\text{CPI}}$

$$\therefore P1 = \frac{3 \times 10^9}{1.5} = 2 \times 10^9 \text{ inst/s}$$

$$P2 = \frac{2.5 \times 10^9}{1} = 2.5 \times 10^9 \text{ inst/s}$$

$$P3 = \frac{4 \times 10^9}{2.2} = 1.82 \times 10^9 \text{ inst/s}$$

$\therefore$ P2 has highest performance.

(b)

Execution time = 10s

$\therefore$ Instructions in 10 seconds
= clock rate $\times 10$

$\therefore$ Instruction executed = $\frac{\text{cycles}}{\text{CPI}}$

For P1:

$$\text{cycles} = 3 \times 10^9 \times 10 = 3 \times 10^{10} \text{ cycles}$$

$$\text{Instructions} = \frac{3 \times 10^{10}}{1.5} = 2 \times 10^{10} \text{ instructions}$$

For P2:

$$\text{cycles} = 2.5 \times 10^9 \times 10 = 2.5 \times 10^{10} \text{ cycles}$$

$$\text{instructions} = \frac{2.5 \times 10^{10}}{1} = 2.5 \times 10^{10} \text{ instructions}$$

For P3:

$$\text{cycles} = 4 \times 10^9 \times 10 = 4 \times 10^{10} \text{ cycles}$$

$$\text{instructions} = \frac{4 \times 10^{10}}{2.2} = 1.82 \times 10^{10} \text{ instructions}$$

(a)

$$T_{\text{new}} = 0.7 T_{\text{old}}$$

$$CPI_{\text{new}} = 1.2 CPI_{\text{old}}$$

We know,

$$T_{\text{new}} = 0.7 T_{\text{old}}$$

$$\Rightarrow \frac{IC \times CPI_{\text{new}}}{f_{\text{new}}} = 0.7 \frac{IC \times CPI_{\text{old}}}{f_{\text{old}}}$$

$$\Rightarrow \frac{CPI_{\text{new}}}{f_{\text{new}}} = \frac{0.7 \times CPI_{\text{old}}}{f_{\text{old}}}$$

$$\Rightarrow f_{\text{new}} = \frac{CPI_{\text{new}} \times f_{\text{old}}}{0.7 CPI_{\text{old}}}$$

$$= \frac{1.2 CPI_{\text{old}} \times f_{\text{old}}}{0.7 CPI_{\text{old}}}$$

$$= \frac{1.2}{0.7} \times f_{\text{old}}$$

$$= 1.71 f_{\text{old}}$$

$$\therefore f_{\text{new}} = 1.71 f_{\text{old}}$$

$$\therefore P1 : f_{old} = 3 \times 10^9 \text{ Hz}$$

$$\therefore f_{new} = 3 \times 10^9 \times 1.71 = 5.14 \times 10^9 \text{ Hz}$$

$$\therefore P2 : f_{old} = 2.5 \times 10^9 \text{ Hz}$$

$$\therefore f_{new} = 2.5 \times 10^9 \times 1.71 = 4.28 \times 10^9 \text{ Hz}$$

$$\therefore P3 : f_{old} = 4 \times 10^9 \text{ Hz}$$

$$\therefore f_{new} = 4 \times 10^9 \times 1.71 = 6.84 \times 10^9 \text{ Hz}$$

Answer to the question no: 4

(a) Given that,

Total execution Time = 250s

Breakdown:

$$\text{Add} = 70s$$

$$\text{sub} = 85s$$

$$\text{Left-shift} = 40s$$

$$\begin{aligned}\text{Other instructions} &= (250 - 70 - 85 - 40) \\ &= 55s\end{aligned}$$

Reduce add time by 20%.

$$\begin{aligned}\therefore \text{New add time} &= 70 - 0.20 \times 70 \\ &= 56s\end{aligned}$$

$$\text{New total time} = 56 + 85 + 40 + 55 = 236$$

$$\text{Time reduced} = 250 - 236 = 14s$$

(b) Target time reduction $= 0.20 \times 250$
 $= 50s$

$\therefore$ New total time must be $= 250 - 50$
 $= 200s$

left shift $= 40s$

Even if left shift becomes 0,
maximum reduction $= 40s$

That gives total time $= 250 - 40$
 $= 210$

So, Not achievable 200s with only left-shift optimization.

Answer to the question no: 5

(a) Given that,

$$P1: \text{clock} = 4 \times 10^9 \text{ Hz}, \text{CPI} = 0.9, I_1 = 5 \times 10^9$$

$$P2: \text{clock} = 3 \times 10^9 \text{ Hz}, \text{CPI} = 0.75, I_2 = 1 \times 10^9$$

$$\begin{aligned} \therefore \text{MIPS } 1 &= \frac{\text{clock rate}}{\text{CPI} \times 10^6} \\ &= \frac{4 \times 10^9}{0.9 \times 10^6} = 4444.44 \text{ MIPS} \end{aligned}$$

$$\begin{aligned} \therefore \text{Execution Time } T_1 &= \frac{I \times \text{CPI}}{\text{clock rate}} \\ &= \frac{5 \times 10^9 \times 0.9}{4 \times 10^9} \\ &= 1.125 \text{ s} \end{aligned}$$

$$\begin{aligned} \therefore \text{MIPS } 2 &= \frac{3 \times 10^9}{0.75 \times 10^6} \\ &= 4000 \text{ MIPS} \end{aligned}$$

$$\begin{aligned} \therefore \text{Execution Time } T_2 &= \frac{I \times \text{CPI}}{\text{clock rate}} \\ &= \frac{1 \times 10^9 \times 0.75}{3 \times 10^9} \\ &= 0.25 \text{ s} \end{aligned}$$

No. largest MIPS does not mean largest performance.

As P1 has the larger MIPS but it takes larger time to execute so performance is measured by execution time, not by MIPS.

Answers to the question no. 6

(a)

$$C_{\text{new}} = (1 - 0.143) C_{\text{old}} \\ = 0.857 C_{\text{old}}$$

$$V_{\text{new}} = 0.813 V_{\text{old}}$$

$$f_{\text{new}} = 0.813 f_{\text{old}}$$

$$\therefore \frac{P_{\text{new}}}{P_{\text{old}}} = \frac{C_{\text{new}} \times V_{\text{new}} \times f_{\text{new}}}{C_{\text{old}} \times V_{\text{old}} \times f_{\text{old}}}$$

$$\Rightarrow \frac{P_{\text{new}}}{P_{\text{old}}} = \frac{0.857 C_{\text{old}} \times (0.813) V_{\text{old}} \times 0.813 f_{\text{old}}}{C_{\text{old}} \times V_{\text{old}} \times f_{\text{old}}}$$

$$\frac{P_{\text{new}}}{P_{\text{old}}} = 0.857 \times (0.813)^2 \times 0.813$$

$$= \frac{P_{\text{new}}}{P_{\text{old}}} = 0.4606$$

$$\therefore \text{Percentage reduction of power} = (1 - 0.4606) \times 100 \\ = 53.94\%$$

Answers to the question no: 7

(a)

From the figure,

$$\text{Clock period } T = 100 \text{ ps}$$

(b)

$$\text{Frequency } f = \frac{1}{T} = \frac{1}{100 \times 10^{-12}} = 10 \text{ GHz}$$

(c)

$$\text{CPI (add)} = 24, \text{ From 100ps to 400ps}$$

$$\text{CPI (sub)} = 23, \text{ from 200ps to 300ps}$$

(d)

$$\text{Average CPI} = \frac{21 \times 4 + 3 \times 3}{21 + 3} = 3.875$$

(e)

$$\text{Execution Time} = \text{Instruction count} \times \text{Avg CPI} \times \text{Clock period}$$

$$= 24 \times 3.875 \times 100$$

$$= 9300 \text{ ps.}$$

(f) SPEC ratio = $\frac{\text{Reference Time}}{\text{Execution Time}}$

$$= \frac{1080}{9360}$$
$$= 0.1161 \underline{\underline{M}}$$

Answer to the question no:02

(a)

Given that,

$$\text{Total instruction} = 1.0 \times 10^6$$

class fractions:-

$$A = 0.30$$

$$B = 0.50$$

$$C = 0.10$$

$$D = 0.10$$

CPI per class:

$$A = 7$$

$$B = 2$$

$$C = 3$$

$$D = 6$$

$$\text{Average CPI} = \sum (\text{fraction}_i \times \text{CPI}_i)$$

$$\therefore \text{CPI for PlayStation 5} = (0.30 \times 7 + 0.50 \times 2 + 0.10 \times 3 + 0.10 \times 6)$$

$$= 4.500 \quad \underline{\text{Ans}}$$

(b) For Xbox Series X :-

CPI per class:

$$A = 5$$

$$B = 4$$

$$C = 2$$

$$D = 1$$

$\therefore$ Average CPI for Xbox Series X :-

$$= 0.30 \times 5 + 0.50 \times 4 + 0.10 \times 2 + 0.10 \times 1$$

$$= 3.850 \quad \underline{\underline{\text{Ans}}}$$

(c) $(4.000 - 3.850) = 0.250$ cycles/instruction

$\therefore 0.250^{\text{more}}$ cycles per instruction on average
does PS5 take vs Xbox.

(d)

Formula:

$$\text{Execution Time} = \frac{\text{Instruction count} \times \text{Average CPI}}{\text{Clock rate.}}$$

For PlayStation 5:-

$$\begin{aligned}\text{clock rate} &= 2.7 \text{ GHz} \\ &= 2.7 \times 10^9 \text{ Hz}\end{aligned}$$

$$\therefore T_{PS5} = \frac{1 \times 10^6 \times 4.000}{2.7 \times 10^9} = 1.481 \text{ ms}$$

For Xbox Series X:

$$\text{clock} = 3 \text{ GHz} = 3 \times 10^9 \text{ Hz}$$

$$\therefore T_{XB} = \frac{1 \times 10^6 \times 3.800}{3 \times 10^9} = 1.267 \text{ ms}$$

$$\begin{aligned}\therefore \text{Difference} &= 1.481 - 1.267 \\ &= 0.215 \text{ ms}\end{aligned}$$

(c) Given that,

Reference time = 120 ms

$$\therefore SPEC_{PSS} = \frac{\text{reference time}}{\text{measured time}}$$

$$= \frac{120}{1.481}$$

$$= 81.03$$

Ans